always be trying to be the intruder in any encounter. They could only achieve this by ceaseless, and otherwise pointless, moving around! Quite apart from the costs in time and energy that would be incurred, this evolutionary trend would, of itself, tend to lead to the category 'resident' ceasing to exist. In a population sitting at the other stable state, 'resident wins, intruder retreats', natural selection would favour individuals who strove to be residents. For each individual, this would mean holding on to a particular piece of ground, leaving it as little as possible, and appearing to 'defend' it. As is now well known, such behaviour is commonly observed in nature, and goes by the name of 'territorial defence'.

The neatest demonstration I know of this form of behavioural asymmetry was provided by the great ethologist Niko Tinbergen, in an experiment of characteristically ingenious simplicity.* He had a fish-tank containing two male sticklebacks. The males had each built nests, at opposite ends of the tank, and each 'defended' the territory around his own nest. Tinbergen placed each of the two males in a large glass test tube, and he held the two tubes next to each other and watched the males trying to fight each other through the glass. Now comes the interesting result. When he moved the two tubes into the vicinity of male *A*'s nest, male *A* assumed an attacking posture, and male *B* attempted to retreat. But when he moved the two tubes into male *B*'s territory, the tables were turned. By simply moving the two tubes from one end of the tank to the other, Tinbergen was able to dictate which male attacked and which retreated. Both males were evidently playing the simple conditional strategy: 'if resident, attack; if intruder, retreat'.

Biologists often ask what the biological 'advantages' of territorial behaviour are. Numerous suggestions have been made, some of which will be mentioned later. But we can now see that the very question may be superfluous. Territorial 'defence' may simply be an ESS which arises because of the asymmetry in time of arrival that usually characterizes the relationship between two individuals and a patch of ground.

Presumably the most important kind of non-arbitrary asymmetry is in size and general fighting ability. Large size is not necessarily always the most important quality needed to win fights, but it is probably one of them. If the larger of two fighters always wins, and if each individual knows for certain whether he is larger or smaller than his opponent, only one strategy makes any sense: 'If your opponent is larger than you, run